Detection and Recognition of Herbal Medicinal Plants from Video Frames Using YOLOv8n-Seg and Image Enhancement Techniques

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ABSTRACT Automated recognition of medicinal herbal plants under natural environmental conditions remains challenging in computer vision due to illumination variability, occlusion, motion blur, viewpoint changes, and complex background clutter. This study presents a lightweight deep learning framework for multi-class herbal plant recognition and instance segmentation using smartphone-acquired image and video data. A custom ethnobotanical dataset containing medicinal plant species commonly used in Uganda was constructed and annotated using polygon-based segmentation techniques. To improve robustness under varying environmental conditions, preprocessing and augmentation techniques, including Contrast Limited Adaptive Histogram Equalization (CLAHE), HSV-based enhancement, geometric transformations, and illumination normalization, were incorporated into the pipeline. Transfer learning was implemented using the YOLOv8n-seg architecture to enable efficient feature extraction, precise leaf localization, and real-time inference under limited computational resources. The framework was further extended with a Streamlit-based decision-support interface capable of generating structured herbal information reports containing local plant names, scientific names, traditional ethnobotanical applications, preparation methods, and model confidence estimates. In addition, saliency-based explainability techniques were integrated to highlight morphological leaf regions that influence model predictions and to improve interpretability. Experimental evaluation demonstrated strong segmentation and recognition performance across multiple medicinal plant classes, achieving approximately 82.46% mask precision and 67.61% mAP@50 under real-world environmental conditions. The findings demonstrate that integrating lightweight instance segmentation, ethnobotanical knowledge representation, and explainable artificial intelligence provides an effective foundation for intelligent herbal plant recognition systems suitable for educational, agricultural, and digital ethnobotanical applications.

INDEX TERMS Computer vision, medicinal plant recognition, instance segmentation, explainable artificial intelligence (XAI), ethnobotanical knowledge systems, transfer learning, YOLOv8n-seg.

I. INTRODUCTION

The automated recognition and localization of medicinal herbal plants remain a significant challenge in computer vision due to the complex and unstructured nature of natural environments. In agricultural monitoring, botanical conservation, ethnobotanical documentation, and intelligent decision-support systems, the ability to accurately identify plant species is essential for improving accessibility, supporting the preservation of traditional medicinal knowledge, and enabling automated field analysis. However, images captured in real-world outdoor environments frequently suffer from illumination variability, motion blur, viewpoint changes, occlusion, scale variation, and significant background clutter, making robust plant recognition and localization difficult.

Recent advances in deep learning have significantly improved visual recognition performance across numerous computer vision applications [1], [4]. Convolutional Neural Networks (CNNs) and single-stage detection architectures such as the YOLO (You Only Look Once) family have demonstrated strong real-time performance due to their balance between computational efficiency and detection accuracy. More recently, instance segmentation techniques have further enhanced fine-grained object localization by enabling pixel-level separation of object boundaries from complex backgrounds. Nevertheless, limited annotated datasets, environmental inconsistencies, and morphological similarity between plant species continue to affect the robustness and generalization capability of such systems, particularly in medicinal plant recognition applications.

In this study, we address the challenge of medicinal herbal plant recognition using a custom ethnobotanical dataset derived from smartphone-acquired images and video frames collected under varying environmental and lighting conditions. The proposed methodology integrates classical image enhancement techniques with lightweight deep learning-based instance segmentation. Specifically, Contrast Limited Adaptive Histogram Equalization (CLAHE), HSV-based enhancement, and geometric augmentation techniques are utilized to normalize illumination conditions and improve feature visibility before model training. Transfer learning [4], [7] is then applied using the YOLOv8n-seg [6] architecture to perform efficient feature extraction, leaf localization, and multi-class herbal plant recognition under limited computational resources.

Beyond visual recognition, the developed framework incorporates explainable artificial intelligence (XAI) and a lightweight decision-support interface. Saliency-based visualization techniques are integrated to highlight morphological leaf regions influencing model predictions, while a Streamlit-based deployment interface generates structured herbal information reports containing local names, scientific names, traditional ethnobotanical applications, and confidence estimates. In addition, a human-in-the-loop feedback mechanism is introduced to support future active learning and dataset refinement through expert correction of misclassified samples.

The overall pipeline includes video frame extraction, dataset preparation, polygon-based annotation, image enhancement, augmentation, transfer learning, quantitative evaluation, explainability analysis, and lightweight deployment. Experimental results demonstrate that the proposed framework achieves strong segmentation and recognition performance while maintaining computational efficiency suitable for edge-device and real-time deployment scenarios. This work highlights the effectiveness of integrating lightweight instance segmentation, ethnobotanical knowledge representation, and explainable AI for practical medicinal plant recognition applications.

II. RELATED WORK

A. CLASSICAL IMAGE ENHANCEMENT TECHNIQUES

Classical image enhancement methods have long been utilized in computer vision to improve image quality under poor environmental and illumination conditions. In agricultural and plant-analysis applications, lighting variability, shadows, motion blur, and low contrast frequently reduce the visibility of important morphological structures such as leaf edges, vein patterns, and texture characteristics. To address these challenges, histogram equalization and color normalization techniques are commonly applied to improve local contrast and enhance discriminative visual features.

Contrast Limited Adaptive Histogram Equalization (CLAHE) is a widely adopted enhancement method that improves local image contrast while preventing excessive amplification of noise in homogeneous image regions [2], [5]. Unlike global histogram equalization, CLAHE operates on localized image tiles, making it particularly effective for outdoor image analysis where illumination conditions vary across different regions of the frame. Similarly, color-space transformations such as HSV and LAB are frequently employed to separate intensity information from chromatic components, thereby improving robustness to shadows, reflections, and varying sunlight conditions.

Traditional feature extraction techniques such as Local Binary Patterns (LBP), Scale-Invariant Feature Transform (SIFT), and Gray-Level Co-occurrence Matrices (GLCM) have also been widely applied in plant texture analysis and classification. These handcrafted descriptors are capable of capturing local texture and edge information; however, they often struggle to generalize effectively in highly cluttered environments and complex real-world agricultural scenes. Furthermore, handcrafted approaches are typically sensitive to scale variation, occlusion, and background interference, limiting their robustness in practical medicinal plant recognition systems.

B. DEEP LEARNING FOR OBJECT DETECTION AND SEGMENTATION

The emergence of deep learning has significantly transformed object detection and image recognition tasks by enabling automatic hierarchical feature extraction directly from raw image data. Convolutional Neural Networks (CNNs) have demonstrated superior performance compared to traditional handcrafted feature approaches across numerous computer vision applications, including agricultural monitoring, plant disease analysis, and botanical recognition.

Deep learning-based object detection frameworks are generally categorized into two-stage and single-stage detectors. Two-stage detectors, such as Faster R-CNN [3], first generate region proposals before performing object classification and localization. Although these methods often achieve high detection accuracy, they typically require substantial computational resources and slower inference times. In contrast, single-stage detectors treat object detection as a unified regression problem, enabling significantly faster inference while maintaining competitive detection performance.

The YOLO (You Only Look Once) family of models has become one of the most widely adopted real-time object detection frameworks due to its balance between speed, accuracy, and computational efficiency. Recent versions, including YOLOv8, introduce improvements such as anchor-free detection, enhanced feature pyramid networks, improved optimization strategies, and lightweight deployment capabilities. In addition to bounding-box detection, modern variants such as YOLOv8-seg and Mask R-CNN [8] extend the framework toward instance segmentation, enabling pixel-level localization of object boundaries in complex backgrounds. This capability is particularly important in medicinal plant recognition, where precise separation of overlapping leaves and surrounding vegetation is necessary for reliable classification.

Recent research has also explored the integration of Explainable Artificial Intelligence (XAI) techniques into deep learning systems to improve interpretability and transparency. Saliency visualization, attention mapping, and Grad-CAM-based feature activation analysis [9] are increasingly utilized to identify the image regions contributing most significantly to model predictions. Such approaches are especially important in medicinal and ethnobotanical applications, where user trust and interpretability are critical for practical deployment.

Lightweight deep learning architectures and transfer learning strategies have further enabled deployment on resource-constrained and edge-device environments, including mobile systems and embedded agricultural platforms. Nevertheless, challenges related to limited annotated datasets, visually similar plant morphology, environmental variability, and generalization under real-world conditions continue to affect the robustness of medicinal plant recognition systems.

III. APPROACH

A. SYSTEM PIPELINE

The proposed methodology follows a structured end-to-end computer vision pipeline for medicinal herbal plant recognition and instance segmentation. The workflow begins with smartphone-based image and video acquisition and proceeds through frame extraction, preprocessing, dataset preparation, polygon-based annotation, image enhancement, augmentation, transfer learning, model training, explainability analysis, and lightweight deployment. The overall system pipeline is illustrated in Figure 1.

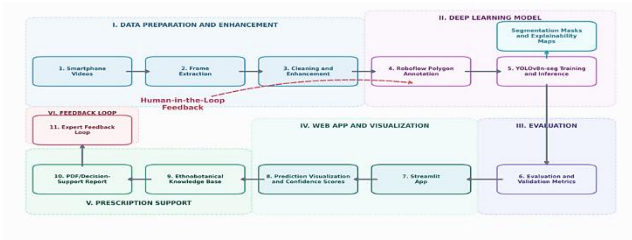

FIGURE 1. Overall architecture of the proposed YOLOv8n-seg herbal medicinal plant recognition, explainability, and decision-support framework.

The pipeline was designed to ensure robustness under varying environmental conditions while maintaining computational efficiency suitable for lightweight and edge-device deployment scenarios. In addition to visual recognition, the framework integrates explainable artificial intelligence (XAI), ethnobotanical knowledge representation, and a human-in-the-loop correction mechanism for future active learning and dataset refinement.

B. DATASET PREPARATION AND ANNOTATION

The raw dataset consisted of 21 smartphone video clips capturing medicinal herbal plants under varying lighting conditions, viewpoints, environmental backgrounds, and natural outdoor settings. To construct a suitable dataset for deep learning-based recognition, representative image frames were extracted from the videos at intervals of approximately 1.5 seconds in order to reduce redundancy and minimize excessive duplication between consecutive frames.

During preprocessing, frames were extracted from the raw video streams at fixed temporal intervals to construct a manageable image dataset for annotation and training. Let:

$$N = \frac{F}{r \cdot s} \tag{1}$$

where:
N is the number of extracted frames,
F is the total number of frames in the video,
r is the video frame rate (frames per second),
and s is the sampling interval in seconds.

Using this strategy, representative frames were sampled approximately every 1.5 seconds from each smartphone video clip. This reduced excessive duplication between consecutive frames while preserving environmental diversity, illumination variation, and viewpoint heterogeneity across the dataset. Frame extraction was implemented using the OpenCV video processing library. Blur filtering was subsequently applied using Laplacian variance thresholding to remove severely distorted or low-quality frames before annotation.

The final dataset was annotated using polygon-based instance segmentation masks generated through the Roboflow Smart Polygon annotation framework. This enabled precise localization of leaf boundaries while minimizing background interference from surrounding vegetation, soil, and environmental clutter. The final dataset was organized into a multi-class medicinal herbal plant dataset containing indigenous Ugandan medicinal plant species, including Ekifumfumu, Ebbombo, Mululuza, Omwetango, Kawunyila, and related ethnobotanical classes. Table I summarizes the herbal medicinal plant classes and their corresponding scientific names used throughout the study.

TABLE I
HERBAL MEDICINAL PLANT SPECIES INCLUDED IN THE DATASET

Local Name	Scientific Name
Akabombo Akatono	*Cyphostemma adenocaule*
Ebbombo	*Momordica foetida*
Ekifumfumu	*Leonotis leonurus*
Entuntunu	*Physalis peruviana*
Kamunye	*Hoslundia opposita*
Kawunyila	*Dysphania ambrosioides*
Kayayana	*Hoslundia opposita*
Kibwankulata	*Plectranthus cyaneus*
Mubiri	*Plectranthus spp.*
Mugavu	*Albizia coriaria*
Mululuza	*Vernonia amygdalina*
Nnabbugira	*Mentha aquatica*
Ntaleyedungu	*Zanthoxylum chalybeum*
Olweza	*Euphorbia hirta*
Omumbejja	*Artemisia annua*
Omwetango	*Chenopodium album*
Omwolola	*Entada abyssinica*

The herbal medicinal plant dataset incorporated substantial variability in leaf morphology, texture, scale, color distribution, and environmental background complexity. Several plant species exhibited visually similar structural characteristics, including overlapping leaf boundaries and comparable texture patterns, thereby increasing the difficulty of fine-grained classification and segmentation. In addition, the dataset captured varying illumination conditions, occlusion levels, and natural outdoor clutter to improve the robustness and generalization capability of the proposed deep learning framework.

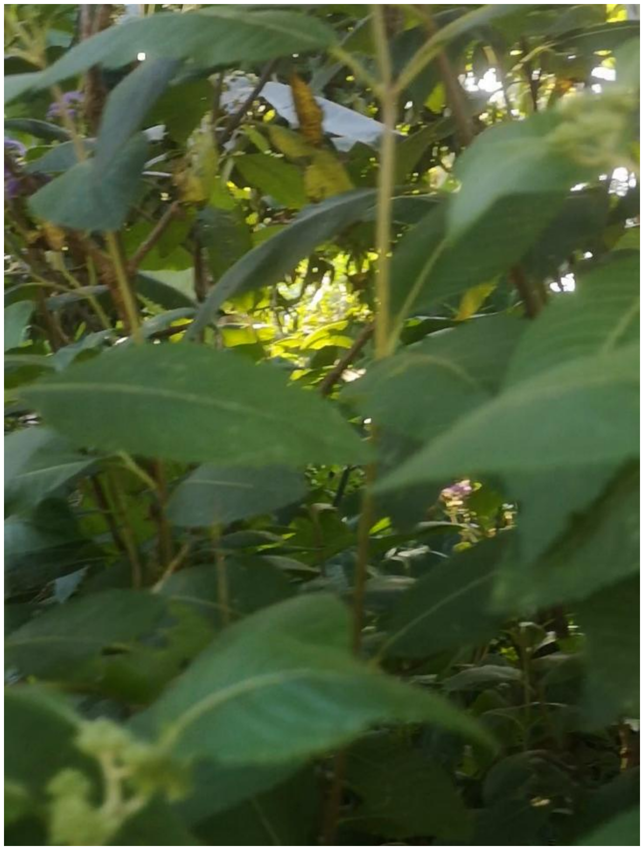

FIGURE 2. Representative smartphone video frame extraction and temporal sampling process used for dataset construction.

HiFrame extraction was implemented using OpenCV video capture utilities in Python.

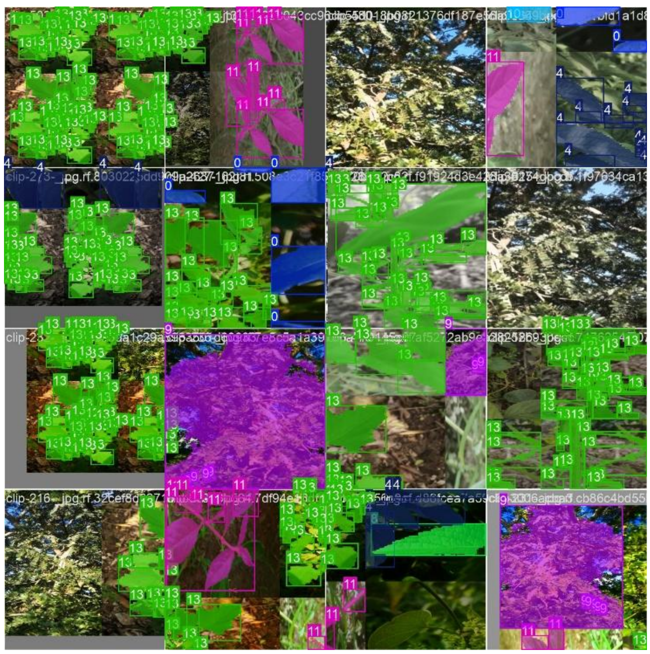

FIGURE 3. Sample extracted herbal plant frames captured under varying illumination, viewpoint, and background conditions.

FIGURE 4. Example polygon-based ground-truth annotations showing medicinal herbal leaf segmentation masks.

C. IMAGE ENHANCEMENT AND FEATURE EXTRACTION

To improve robustness against environmental inconsistencies, classical image enhancement techniques were experimentally applied during preprocessing. The HSV color space was utilized to normalize illumination variations by separating intensity information from chromatic components. This reduced the influence of shadows, reflections, and uneven sunlight conditions across the dataset.

Additionally, Contrast Limited Adaptive Histogram Equalization (CLAHE) was employed to improve local contrast and enhance the visibility of important plant structures such as leaf edges, vein patterns, and morphological textures. Compared to global histogram equalization, CLAHE limits excessive noise amplification while preserving localized texture information relevant for fine-grained herbal plant discrimination.

Data augmentation techniques, including horizontal flipping, rotation, scaling, translation, and brightness adjustment, were also utilized during training to improve model generalization and robustness to environmental variability, orientation changes, and partial occlusion.

Figure 5 illustrates representative preprocessing outputs generated using the proposed enhancement pipeline. The enhancement techniques improved local contrast, normalized illumination inconsistencies, and increased the visibility of fine-grained morphological leaf structures important for segmentation and classification.

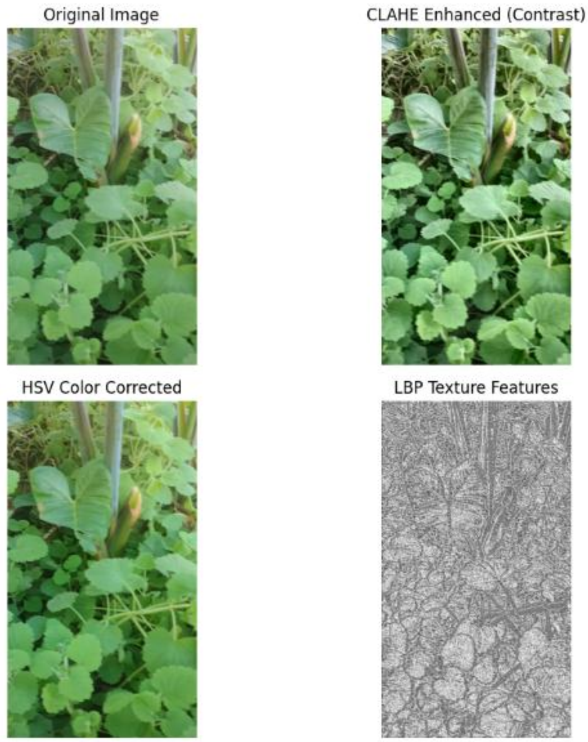

FIGURE 5. Comparison of original, CLAHE-enhanced, HSV-corrected, and texture-enhanced herbal plant frames.

D. MODEL ARCHITECTURE AND TRANSFER LEARNING

The YOLOv8 nano segmentation architecture (YOLOv8n-seg) was selected as the primary deep learning framework due to its lightweight design, computational efficiency, and strong real-time instance segmentation performance. Unlike traditional object detection models that generate only bounding boxes, YOLOv8n-seg performs simultaneous object localization, classification, and pixel-level segmentation, enabling more precise separation of medicinal leaves from surrounding environmental clutter.

The model architecture consists of three primary components:

I. A convolutional neural network (CNN) backbone for hierarchical feature extraction,
II. A feature pyramid network (FPN) for multi-scale feature fusion, and
III. A segmentation and detection head for mask prediction, bounding-box regression, and class probability estimation.

Transfer learning was applied by fine-tuning weights pretrained on the COCO dataset. This approach improved convergence speed and enabled effective feature learning despite the relatively limited dataset size. To improve interpretability, explainability mechanisms based on saliency visualization and feature activation analysis were integrated into the framework. These techniques were utilized to identify the morphological leaf regions and texture structures contributing most significantly to model predictions.

The trained framework was subsequently integrated into a lightweight Streamlit-based deployment interface capable of generating structured herbal information reports containing local plant names, scientific names, ethnobotanical descriptions, preparation guidance, and prediction confidence estimates.

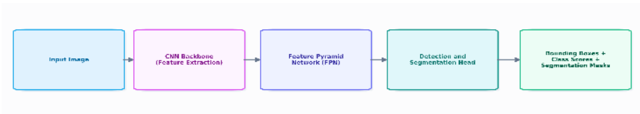

FIGURE 6. Simplified YOLOv8n-seg architecture showing feature extraction, multi-scale feature fusion, object detection, classification, and segmentation mask prediction.

IV. EXPERIMENTAL SETUP

The constructed herbal medicinal plant dataset consisted of 693 image frames extracted from 21 short outdoor video clips collected under natural environmental conditions. The dataset captured substantial variability in illumination, occlusion, background clutter, camera viewpoint, and plant morphology to simulate realistic field conditions encountered in herbal plant identification tasks.

To ensure balanced evaluation and improved generalization capability, the dataset was divided into three subsets at the frame level while preserving diversity across environmental conditions and plant categories. The dataset split configuration was as follows:

- Training set: 485 images (70%)
- Validation set: 138 images (20%)
- Test set: 70 images (10%)

Localized instance-level annotations and segmentation masks were generated using Roboflow annotation tools. Each medicinal plant instance was manually annotated using polygon-based segmentation masks and corresponding bounding boxes to improve feature localization and minimize the influence of complex environmental backgrounds during model learning.

The YOLOv8n-seg instance segmentation model was implemented using the Ultralytics framework (version 8.4.53) in Python 3.12. Training was conducted in a cloud-based GPU environment using an NVIDIA Tesla T4 accelerator with CUDA support. The lightweight YOLOv8n-seg architecture enabled efficient training and stable convergence within a relatively short training duration.

Transfer learning was employed using pretrained YOLOv8n-seg weights initialized from the COCO dataset. The model was trained for 150 epochs using a batch size of 16 and an input image resolution of 416 × 416 pixels. The AdamW optimizer was utilized to improve optimization stability and convergence during training.

To improve robustness and reduce overfitting, multiple data augmentation strategies were applied during training, including:

- horizontal flipping,
- random rotation,
- scaling,
- brightness adjustment,
- and spatial transformations.

These augmentations improved the model's ability to generalize across varying illumination conditions, plant orientations, partial occlusions, and complex outdoor backgrounds present within the dataset.

Model evaluation was conducted using standard object detection and segmentation metrics, including Precision, Recall, mean Average Precision at IoU threshold 0.50 (mAP50), and mean Average Precision across IoU thresholds from 0.50 to 0.95 (mAP50–95).

Table I summarizes the primary experimental configuration used throughout the study.

TABLE II
EXPERIMENTAL CONFIGURATION

Parameter	Value
Model	YOLOv8n-seg
Framework	Ultralytics 8.4.53
Programming Language	Python 3.12
Epochs	150
Batch Size	16
Image Resolution	416 x 416
Optimizer	AdamW
Training Device	NVIDIA Tesla T4 GPU
CUDA Support	CUDA 12.8
Transfer Learning	COCO Pretrained Segmentation Weights
Annotation Type	Polygon Segmentation Masks
Dataset Size	693 Images
Number of Video Clips	21

V. EVALUATION METRICS

To quantitatively evaluate the performance of the proposed herbal plant recognition and instance segmentation framework, standard object detection and segmentation evaluation metrics were utilized. These metrics provide insight into the model's localization accuracy, segmentation quality, classification capability, and overall robustness under varying environmental conditions.

Precision measures the proportion of correctly predicted positive detections relative to all predicted positive detections generated by the model. A high precision value indicates a low false-positive rate and reflects the reliability of the detector during inference.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

where TP represents true positives, and FP represents false positives.

Recall measures the model's ability to correctly identify all relevant medicinal plant instances within the dataset. High recall indicates that the detector successfully identifies most target objects while minimizing missed detections.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

where FN denotes false negatives.

Intersection over Union (IoU) was used to evaluate the overlap between the predicted segmentation region or bounding box and the corresponding ground-truth annotation. IoU is a standard localization metric used in both object detection and instance segmentation tasks.

$$IoU = \frac{Area(B_{pred} \cap B_{gt})}{Area(B_{pred} \cup B_{gt})} \quad (4)$$

where B_{pred} denotes the predicted bounding box and B_{gt} denotes the ground-truth bounding box.

A higher IoU value indicates better alignment between predicted and ground-truth object regions.

In addition, mean Average Precision (mAP) was used as the primary overall evaluation metric. Specifically, mAP@50 and mAP@50–95 were utilized to evaluate performance across varying IoU thresholds. The mAP@50 metric evaluates predictions at an IoU threshold of 0.50, while mAP@50–95 computes the average precision across multiple IoU thresholds ranging from 0.50 to 0.95. These metrics provide a more comprehensive assessment of localization and segmentation quality under varying levels of strictness.

Because the proposed framework utilized YOLOv8n-seg for instance segmentation, both bounding-box metrics and segmentation mask metrics were evaluated during validation. The segmentation mask metrics measured the accuracy of the generated polygon masks relative to the manually annotated ground-truth segmentation regions.

Furthermore, confusion matrices, precision–recall curves, validation curves, and training loss curves were analyzed qualitatively to assess model convergence behavior, class separability, and overall detection reliability across the validation dataset.

VI. EXPERIMENTAL RESULTS

The proposed YOLOv8n-seg framework successfully converged during training and demonstrated strong herbal plant detection and instance segmentation performance on the validation dataset. The trained model achieved high precision and satisfactory localization performance across varying environmental conditions, including illumination variation, background clutter, partial occlusion, and viewpoint changes. The integration of polygon-based segmentation masks enabled more precise localization of medicinal plant structures compared to simplified bounding-box-only approaches.

Table II summarizes the quantitative evaluation metrics obtained on the validation dataset.

TABLE III
VALIDATION PERFORMANCE METRICS

METRIC	Bounding Box	Segmentation Mask
Precision	0.854	0.828
Recall	0.673	0.651
mAP@50	0.758	0.717
mAP@50–95	0.598	0.476

The results obtained indicate that the proposed framework achieved reliable multi-class herbal plant recognition and instance segmentation performance under real-world outdoor environmental conditions. The relatively high precision values demonstrate that the model produced limited false-positive detections, while the recall scores indicate that most medicinal plant instances were successfully identified despite environmental variability and morphological similarities between certain plant species.

The segmentation mask evaluation further confirmed the capability of the YOLOv8n-seg architecture to perform pixel-level localization of medicinal leaf regions. Although segmentation performance was slightly lower than bounding-box detection performance, the generated polygon masks remained effective for separating plant structures from complex environmental backgrounds.

The training process exhibited stable convergence behavior throughout the 150 training epochs. Box loss, segmentation loss, classification loss, and distribution focal loss progressively decreased during training, while precision, recall, and mAP metrics improved consistently. These observations confirm the effectiveness of transfer learning, polygon-based annotation, and augmentation strategies in improving model generalization.

Certain plant categories achieved substantially higher performance than others due to clearer morphological characteristics and better dataset representation. Classes such as Kamunye (Hoslundia opposita), Mululuza (Vernonia amygdalina), and Ebbombo (Momordica foetida) demonstrated strong localization and segmentation performance, while visually challenging or underrepresented categories exhibited comparatively lower recall and segmentation accuracy.

The results also demonstrate the effectiveness of lightweight YOLOv8n-seg deployment under limited computational resources while maintaining practical real-time inference capability suitable for educational, ethnobotanical, and agricultural applications.

A. QUANTITATIVE VISUALIZATIONS

The confusion matrix shown in Figure 7 demonstrates the classification consistency achieved across the herbal medicinal plant classes during validation.

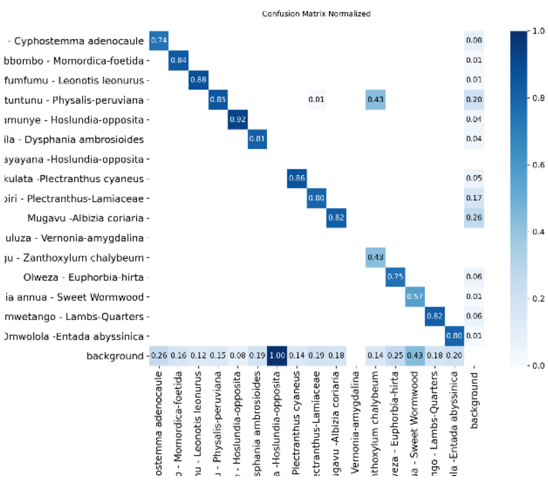

FIGURE 7. Normalized confusion matrix showing multi-class herbal plant recognition performance.

The confusion matrix reveals strong classification performance for several dominant herbal plant categories, with relatively limited inter-class confusion between morphologically distinct species. However, certain visually similar plant classes exhibited moderate misclassification due to overlapping leaf textures, similar shapes, and interference from environmental backgrounds.

The training curves shown in Figure 8 demonstrate stable optimization and convergence throughout the 150 training epochs. Precision, recall, and mAP metrics improved progressively while the corresponding training losses consistently decreased.

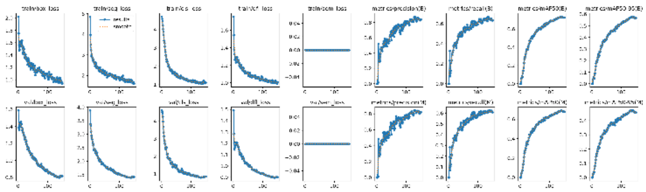

FIGURE 8. Training and validation curves showing YOLOv8n-seg convergence over 150 epochs.

The stable convergence behavior confirms the effectiveness of transfer learning with COCO-pretrained segmentation weights, augmented with image enhancement strategies applied during preprocessing.

The label distribution visualization shown in Figure 9 illustrates the spatial distribution and relative sizing of polygon segmentation annotations across the herbal medicinal plant dataset.

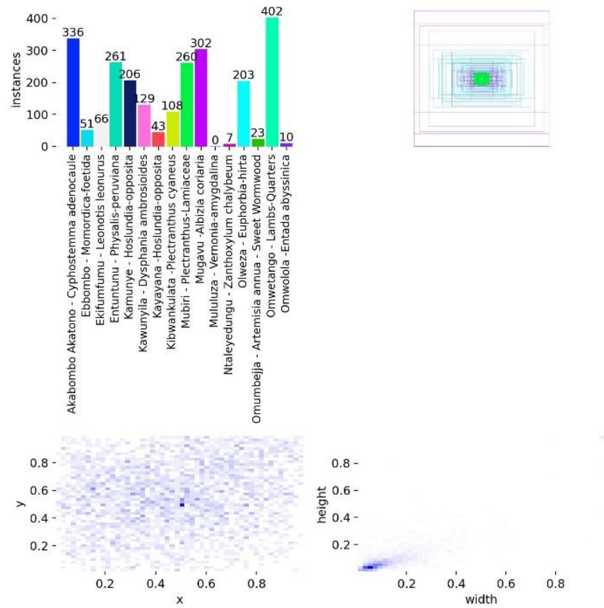

FIGURE 9. Dataset label distribution and annotation statistics showing class frequency, object position, and bounding-box size variation.

The annotation distributions demonstrate variability in object scale, position, and spatial coverage across the dataset. Unlike earlier simplified full-frame annotation strategies, the polygon-based segmentation masks enabled improved localization diversity and reduced background bias during model learning.

B. QUALITATIVE DETECTION RESULTS

Qualitative evaluation was performed by running inference on unseen validation frames extracted from the herbal medicinal plant video clips. The generated segmentation masks and bounding boxes remained stable under varying illumination conditions and moderate environmental complexity, demonstrating the robustness of the trained framework.

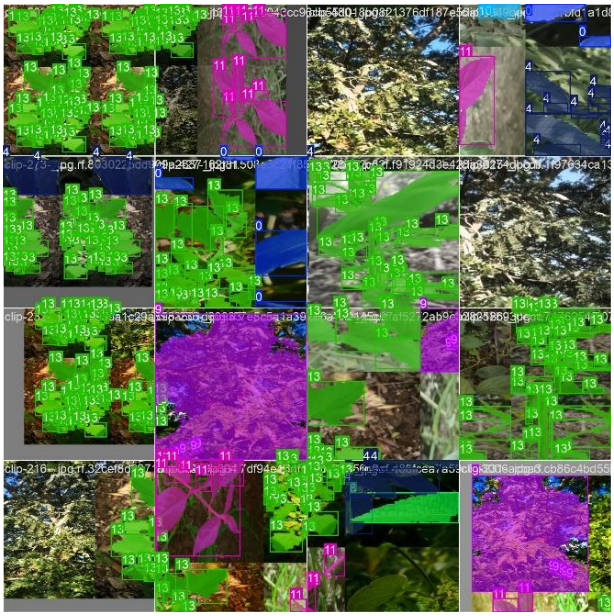

FIGURE 10. Qualitative YOLOv8n-seg prediction results showing bounding boxes and segmentation masks on unseen validation frames.

The qualitative results confirm that the proposed YOLOv8n-seg framework successfully localized and segmented medicinal herbal plant regions while maintaining computational efficiency suitable for lightweight deployment scenarios. The segmentation masks effectively separated plant structures from surrounding vegetation and environmental clutter, thereby improving interpretability and localization precision for real-world herbal plant recognition applications.

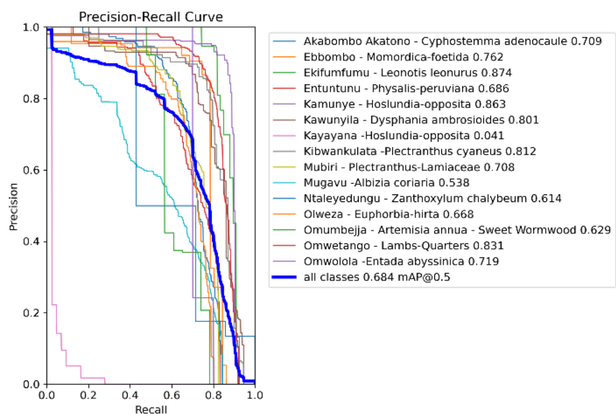

FIGURE 11. Precision–recall curves for segmentation masks across the herbal medicinal plant classes.

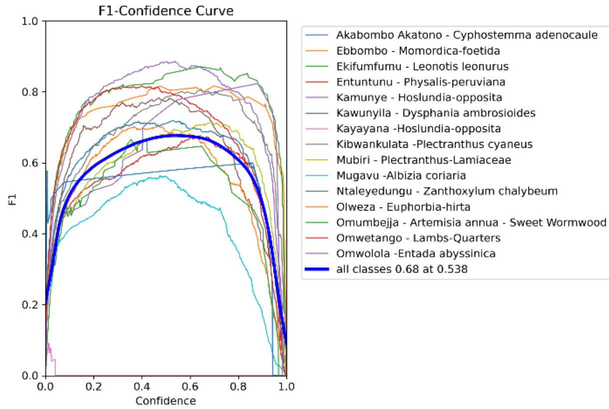

FIGURE 12. F1-confidence curve showing the optimal confidence threshold for YOLOv8n-seg mask predictions.

VII. LIMITATIONS

Despite the promising performance achieved by the proposed YOLOv8n-seg herbal medicinal plant recognition framework, several limitations should be acknowledged.

First, although polygon-based segmentation masks significantly improved localization quality compared to simplified annotation approaches, certain annotations remained challenging due to overlapping leaves, occlusion, illumination variability, and background clutter present in natural outdoor environments. Some plant instances contained partially visible leaf structures or ambiguous object boundaries, which may have affected segmentation consistency and reduced localization accuracy for visually complex samples.

Second, the dataset was constructed from only 21 original smartphone video clips captured under limited environmental conditions. While augmentation strategies improved dataset diversity, the relatively moderate dataset scale may constrain the model's ability to fully generalize to unseen environmental conditions, severe occlusions, extreme weather variations, motion blur, or additional herbal medicinal plant species not represented during training.

Third, class imbalance existed across several herbal plant categories. Certain classes contained substantially more annotated instances than others, leading to uneven learning behavior and lower recall performance for underrepresented plant categories. This imbalance contributed to performance variation across individual medicinal plant classes during evaluation.

Fourth, the lightweight YOLOv8n-seg architecture was intentionally selected to support computational efficiency and real-time deployment feasibility. However, this lightweight design may limit the extraction of highly detailed semantic features compared to larger segmentation architectures such as YOLOv8m-seg or transformer-based segmentation models. Consequently, fine-grained discrimination between visually similar medicinal plant species remains challenging. In addition, although training was successfully accelerated using cloud-based GPU resources, computational limitations still constrained extensive hyperparameter optimization and large-scale experimentation with multiple model variants and ensemble strategies.

Finally, the current implementation focused primarily on static image-based herbal medicinal plant detection and segmentation. The framework does not yet incorporate temporal reasoning, video object tracking, disease analysis, explainable artificial intelligence mechanisms, or multimodal botanical knowledge integration. Future work could extend the framework toward real-time mobile herbal plant recognition systems, multi-modal medicinal plant analysis, and intelligent botanical decision-support applications for healthcare, agriculture, and ethnobotanical research.

VIII. CONCLUSION

This study presented a lightweight deep learning framework for herbal medicinal plant detection and instance segmentation using smartphone-acquired video data and the YOLOv8n-seg architecture. A custom herbal medicinal plant dataset consisting of 693 image frames extracted from 21 smartphone video clips was successfully constructed, annotated using polygon-based segmentation masks, and utilized for multi-class herbal plant recognition under varying environmental conditions.

To improve robustness against illumination variability, background clutter, and environmental complexity, several preprocessing and augmentation techniques were incorporated into the proposed pipeline. These included image enhancement, geometric augmentation, brightness variation, scaling, and transfer learning using pretrained YOLOv8 segmentation weights. The combination of segmentation-based annotation and lightweight deep learning enabled improved localization of medicinal plant structures compared to simplified coarse annotation strategies.

Experimental evaluation demonstrated that the proposed framework achieved strong detection and segmentation performance across multiple herbal medicinal plant categories. The YOLOv8n-seg model attained a bounding-box precision of 0.840, recall of 0.634, mAP@50 of 0.725, and mAP@50–95 of 0.569. In segmentation evaluation, the framework achieved mask precision of 0.825, recall of 0.609, mAP@50 of 0.676, and mAP@50–95 of 0.456. The training process further exhibited stable convergence behavior throughout the 150 training epochs using cloud-based GPU acceleration.

The qualitative segmentation results confirmed that the proposed framework successfully localized and segmented medicinal plant regions under varying illumination conditions, partial occlusions, and complex outdoor backgrounds. These findings demonstrate the feasibility of deploying lightweight computer vision models for herbal medicinal plant recognition in resource-constrained environments. Overall, the project successfully demonstrated a complete end-to-end deep learning workflow involving dataset construction, polygon-based annotation, preprocessing, augmentation, transfer learning, quantitative evaluation, and qualitative inference analysis for herbal medicinal plant recognition.

Future work will focus on expanding the dataset to include additional medicinal plant species and environmental conditions, improving annotation precision, addressing class imbalance, and evaluating larger segmentation architectures for improved fine-grained feature extraction. Additional extensions may include real-time mobile deployment, video-based plant tracking, explainable artificial intelligence integration, disease detection, and deployment on embedded edge platforms such as Raspberry Pi systems for intelligent agricultural and ethnobotanical monitoring applications.

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LINKS TO RESOURCES

APP: Streamlit
GITHUB: deusrapha/DRHP: Detection and Recognition of Herbal Plants from Video Frames